

MAINTENANCE CASE FILE

PART 1 — DIAGNOSIS

Activity 05 — Maintaining Your Own System · Day 1

Expediente de mantenimiento — Parte 1: diagnóstico

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Grade and section / Grado y sección	3° Desarrollo de Software "A"
System / Sistema	Entre Estrellas - Cafetería y Souvenirs
Stack / Tecnología	PHP
Current version / Versión	v1.0
Date / Fecha	2026-09-09
Teacher / Docente	Ing. Javier Arturo García Mineros

1. Configuration inventory

1. Inventario de configuración

ITEM	DESCRIPTION (ENGLISH)
Purpose of the system and users	Point-of-sale and back-office system for a coffee shop and souvenir store. Three roles: Administrator (full access), Cashier (sales and read-only inventory), Warehouse/Bodega (inventory and suppliers).
Language, framework and versions	PHP 8 (procedural, no framework), PDO with the sqldr driver for database access, vanilla JavaScript and CSS on the client side.

Database engine and version	Microsoft SQL Server (Express edition), connected through PDO::sqlsrv.
Number of tables and approximate size	10 tables (carrito_tiempo_real, empleados_nomina, inventario_productos, login, pedido_kiosco_live, pedidos_kiosco, productos, proveedores, reportes_estadisticas, ventas). Size: 16.00 MB.
Three largest tables and their record counts	ventas, productos, and pedidos_kiosco, as they store sales transactions, catalog items, and kiosk orders. Exact counts: ventas (22 rows), productos (41 rows), pedidos_kiosco (3 rows).
Dependencies and external services	Google Fonts CDN (Poppins). No payment gateway, no external API. Passwords hashed locally with PHP's native password_hash()/password_verify() (bcrypt) — no third-party auth service.
Where the system runs	Local development machine, Apache via XAMPP (htdocs/entre_estrellas), against a local SQL Server Express instance. Not deployed to a public server yet.
Configuration files (names only, no passwords)	config/conexion.php (server name and database name), config/auth.php (session/role-guard helper).
Credentials custody (who knows what)	SQL Server instance access: known by the developer only (Windows-integrated connection, no password embedded in code). Application-level user passwords: stored as bcrypt hashes in the login table; nobody, including the developer, can read them in plain text.
Location of the official source code	Single local folder (htdocs/entre_estrellas) on the developer's machine. No version control (Git) yet — see finding F-02.
Existing documentation	LEEME.md at the project root: installation steps, test credentials, and a changelog of what was fixed/added in each delivery.
Personal data stored	Personal data stored includes employees' full names, identity document numbers (DUI), phone numbers, email addresses, job titles, and salary records stored in empleados_nomina, as well as usernames and encrypted password hashes in login.

2. Backup and restore evidence

2. Respaldo y prueba de restauración

ITEM	FILE NAME AND DATE	SIZE	WHERE IT IS STORED
Source code backup	entre_estrellas_backup_2026-08-09.zip	61.1 MB	Local directory: C:\xampp\htdocs\
Database export	EntreEstrellasDB.bak / (2026-07-18)	5.84 MB (5,984 KB)	Local disk backup folder / SSMS default path
Second copy location	entre_estrellas_backup_2026-08-09.zip EntreEstrellasDB.bak	66.94 MB	External USB Flash Drive (Kingston USB 3.0)
Restore test performed	EntreEstrellasDB_DEV (Restored Database)	16.00 MB	Local SQL Server Express instance (DESKTOP-AMT4TK3\SQLEXPRESS)

2.1 Restore test

The database was restored under a new name, EntreEstrellasDB_DEV, using SQL Server's RESTORE DATABASE command with WITH MOVE to relocate the data and log files. A separate copy of the application folder was pointed to this restored database by changing the \$baseDatos value in config/conexion.php. Opening panel_control.php and each module confirmed that the same employees, products, suppliers and sales history loaded correctly, which proves the backup file is complete and restorable.

La base de datos se restauró bajo un nombre nuevo, EntreEstrellasDB_DEV, usando el comando RESTORE DATABASE de SQL Server con WITH MOVE para reubicar los archivos de datos y de log. Una copia aparte de la carpeta del proyecto se apuntó a esta base restaurada cambiando el valor \$baseDatos en config/conexion.php. Al abrir panel_control.php y cada módulo se confirmó que cargaban los mismos empleados, productos, proveedores e historial de ventas, lo que prueba que el archivo de respaldo está completo y es restaurable.

2.2 Rollback procedure

1. Stop using the production copy.
2. Restore the latest .bak file over EntreEstrellasDB with WITH REPLACE.
3. Redeploy the last known-good .zip of the source code over htdocs/entre_estrellas, overwriting any in-progress change.

4. Reload public/index.php and verify login works for all three roles (Administrator, Cashier, Warehouse) before announcing the system is back.

1. Dejar de usar la copia de producción.
2. Restaurar el último .bak sobre EntreEstrellasDB con WITH REPLACE.
3. Volver a desplegar el último .zip de código bueno conocido sobre htdocs/entre_estrellas, sobrescribiendo cualquier cambio a medias.
4. Recargar public/index.php y verificar el login de los tres roles (Administrador, Cajero, Bodega) antes de avisar que el sistema volvió a la normalidad.

Attach the evidence: screenshots of the backup files with their dates and sizes, and of the restored copy running.

Adjunte las capturas de los archivos de respaldo con fecha y tamaño, y de la copia restaurada funcionando.

3. Baseline metrics

3. Métricas de línea base

METRIC	VALUE BEFORE	HOW IT WAS MEASURED
Load time of the heaviest screen	1.25 s (average)	Measured via Chrome DevTools Network tab reloading modulos/ventas/ventas_pos.php
Time of the slowest query or report	0.95 s	Measured via Chrome DevTools Network tab when executing modulos/reportes/reportes.php.
Database size	16.00 MB	Checked via SSMS Disk Usage report for EntreEstrellasDB (Data: 8.00 MB, Log: 8.00 MB).
Records in the main table	22 records (dbo.ventas)	Executed SELECT COUNT(*) FROM dbo.ventas; in SQL Server Management Studio.
Known open defects	2 open findings	Counted active corrective maintenance items identified in Section 4 (F-01 and F-06).
Repeated errors in the log	0 (None)	Inspected C:\xampp\apache\logs\error.log; no repetitive PHP or PDO exceptions registered.

Active users	4 users	Executed SELECT COUNT(*) FROM dbo.login; in SQL Server Management Studio.
Date of the last backup before today	2026-09-09 19:37:36	Query executed on msdb.dbo.backupset for EntreEstrellasDB.

4. Findings register

4. Registro de hallazgos

ID	FINDING (ENGLISH)	AREA	SEVERITY	EVIDENCE / HOW YOU DETECTED IT	MAINTENANCE TYPE
F-01	The Cashier role can currently create, edit and permanently delete products (name, price, stock), which is more access than the role needs and risks accidental or malicious catalog changes.	Security	High	Reviewed requerirRol() calls in <code>modulos/inventario/*.php</code> ; Cashier is currently allowed into <code>guardar_producto.php</code> and <code>eliminar_producto.php</code> .	Corrective
F-02	The source code has no version control (no Git repository). Every delivery is a manual folder copy, so there is no history of changes and no easy rollback to a specific point.	Documentation / Process	Medium	Verified: no .git folder exists in the project directory.	Preventive
F-03	The entire user interface is in Spanish only, which does not match the bilingual/English requirement now being added to the project.	Usability	Medium	Visual inspection of all screens (login, dashboard, POS, inventory, kiosk).	Adaptive
F-04	No automated or scheduled backup exists; every backup so far has been manual and only	Preventive	High	Confirmed: no backup job is configured in SQL Server Agent.	Preventive

	tested once, during this diagnosis.				
F-05	Product images are never deleted from public/uploads/productos/ when a product's image is replaced, so the upload folder grows indefinitely with orphaned files.	Storage	Low	Traced guardar_producto.php: it saves a new file but never calls unlink() on the previous imagen_url.	Preventive
F-06	Kiosk orders (pedidos_kiosko) that a cashier never attends stay in "Pendiente" status forever; there is no expiration or cancellation after a timeout.	Functional	Medium	Read the kiosk/POS flow: a pending order has no auto-cancel logic.	Corrective
F-07	There are no automated tests (unit or integration) for critical logic such as stock deduction on sale or password verification.	Preventive	Medium	Reviewed the project structure: only manual php -l syntax checks exist, no test suite.	Preventive
F-08	Reports and inventory tables have no pagination; as detalle_ventas and movimientos_inventario grow, these screens will slow down noticeably.	Performance	Low	Code review of reportes.php and inventario.php: all rows are fetched with fetchAll(), no LIMIT/OFFSET.	Perfective
F-08	Only inventory stock changes are logged (movimientos_inventario); there is no audit trail for who edited or deactivated an employee, a supplier, or reversed a sale.	Security	Low	Compared what each module writes to the database versus what it should log for accountability.	Preventive

4.1 Overall state of the system

Entre Estrellas is a working, database-backed point-of-sale system for a coffee shop, built in PHP against SQL Server. It has been running since its first delivery and has already gone through one full redesign (visual theme, self-service kiosk, printable receipt, and image uploads). Role-based access exists and mostly works, but the Cashier role currently has more permission over the product catalog than it should — this is the highest-severity open finding and the first correction planned for this maintenance cycle, together with translating the interface to English. The system has never had an automated backup, has no version control, and has no automated tests, which makes it functional but fragile: it depends entirely on manual discipline to stay safe while it is being maintained.

Server, construido en PHP para una cafetería. Ha estado en funcionamiento desde su primera entrega y ya pasó por un rediseño completo (tema visual, kiosk de autoservicio, ticket imprimible y subida de imágenes). El control de acceso por rol existe y funciona en su mayoría, pero el rol de Cajero tiene actualmente más permisos sobre el catálogo de productos de los que debería tener — este es el hallazgo abierto de mayor severidad y la primera corrección planeada para este ciclo de mantenimiento, junto con la traducción de la interfaz a inglés. El sistema nunca ha tenido un respaldo automatizado, no tiene control de versiones y no tiene pruebas automatizadas, lo que lo hace funcional pero frágil: depende por completo de la disciplina manual para mantenerse seguro mientras se le da mantenimiento.

Declaration / Declaración

I declare that the data in this file was obtained from my own working system, that the backup was really created and restored, and that no password has been reproduced in this document.

Declaro que los datos provienen de mi propio sistema en funcionamiento, que el respaldo fue creado y restaurado realmente y que no se reprodujo ninguna contraseña en este documento.

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